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COMPLEX ENGINEERING PROBLEM

ELECTROMAGNETIC FIELD THEORY

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ELECTROMAGNETIC FIELD THEORY
EE-241

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Complex Engineering Problem

Design a microstrip Inductor with a center frequency of 2.4 GHz and a bandwidth of 100 MHz using a microstrip transmission line. The substrate material used is Rogers RO4003C with a dielectric constant of 3.38 and a thickness of 0.5 mm. The characteristic impedance of the transmission line is 50 ohms. Design the inductor using a short circuit transmission line model and analyze its performance using an electromagnetic simulation tool. (Hint: use Line Calc to find length and width). Inductance is 100 □H.

Tools Used

- HFSS Ansys software
- Online microstrip calculator

Data

- Impedance = $Z_O = 50\Omega$
- Central frequency = f_c = 2.4GHz
- Bandwidth frequency = f_b = 100MHz
- Substrate= Rogers RO4003C
- Dielectric constant = ε_r = 3.38
- Thickness/ Height = h = 0.5mm
- Inductance = L = 100□H

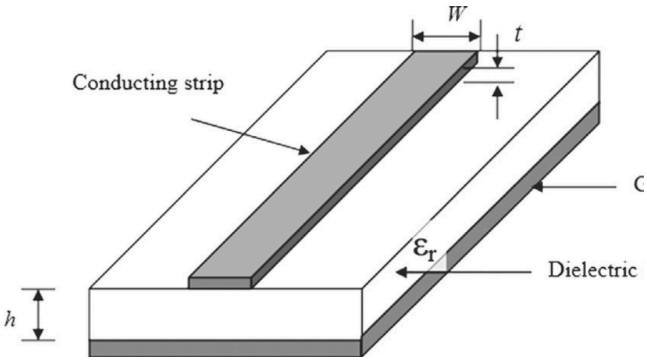
Required

- Length of inductor= *l* =?
- Width of inductor = *w* =?

Introduction

A microstrip is a flat conductor separated by insulating dielectric from a large conducting ground plane. The microstrip is designed so that its length is one-quarter or one-half of the wavelength of the signal propagating through it. Microstrips arose because, at very

high frequencies, the required circuit inductance and capacitance values were too small for standard inductors and capacitors. As such, microstrips serve the same function as inductors and capacitors only at microwave frequencies. They are used as tuned circuits, impedance-matches, filters, phase shifters, and reactive components.



Calculation

For designing a microstrip inductor with specifications as given in the question, we need to find length and width of the inductor as well.

There are different formulae for calculating the length and height, but we will go with the following.

Width of the microstrip

For $Z_0 > 44 - 2\varepsilon_r$:

$$p = \frac{Z0\sqrt{2(\varepsilon r+1)}}{\sqrt{119.9}} + \left[\frac{1(\varepsilon_r - 1)}{2(\varepsilon_r + 1)} \times \left(\ln\frac{\pi}{2} + \frac{1}{\ln\frac{4}{\pi}}\right)\right]$$

here ε_r = 3.38 and Z_o = 50Ω

putting in values in the equation, we get

p = 1.37

and width to thickness, s ratio:

$$s = \frac{w}{h} = \frac{8e^p}{(e^{2p} - 2)}$$

where p =1.37, putting in value:

$$\frac{w}{h} = 2.33$$

Where h= 0.5mm

w = 2.33 × h = 2.33× 0.5mm

Then width is:

***w* = 1.165 mm**

Length of the microstrip

For the length of the microstrip, we need to calculate phase constant, phase velocity, and effective relative permittivity as well.

- Effective relative permittivity (ε_r, eff)

$$\varepsilon_r, eff = \frac{\varepsilon_r + 1}{2}$$

Putting $\varepsilon_r, eff = 3.38$,

$\varepsilon_{eff=}$ 2.19

- Phase velocity

$$v_{p=} \frac{c}{\sqrt{\varepsilon_r, eff}}$$

c = 3×10^8 and $\varepsilon_{eff=}$ 2.19, we get

$v_{p=0.675c}$ m/s

- Phase constant

ß = $\frac{\omega}{v_p}$ = $\frac{2\pi f}{v_p}$

here f =f_c = 2.4GHz

then

ß = 75.02 rad/m

Length of microstrip

For short circuit, formula for length is:

$$l = \frac{1}{ß} tan^{-1}(\frac{\omega L_{eq}}{Z_o})$$

$$l = \frac{1}{ß} tan^{-1}(\frac{2\pi f_c L_{eq}}{Z_o})$$

where f_c = 2.4GHz,

L_{eq} = 100H and ß = 75.02 rad/m

Then length is:

l = 0.02049 m = 20.5 mm

Losses in microstrip

In ideal conditions, there will be no loss in the microstrip but in real,

1. The conductivity of the metal, the skin effect, and surface roughness affect conductor losses in microstrip lines.
2. The dielectric substrate loss tangent drives microstrip line dielectric losses.

3. Radiation losses depend on the dielectric constant, the thickness of the dielectric substrate, and the microstrip line geometry but are often overlooked when compared to ohmic losses and dielectric losses.
4. When circuit designers use magnetic materials, such as ferrites or garnets, as dielectric material, magnetic losses can occur in the circuit.

Mathematically, the dielectric constant and losses on a microstrip line are defined using an effective dielectric constant.

$$\varepsilon_{eff} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + 12\frac{h}{W}\right)^{-1/2} \quad \text{for } \frac{W}{h} \geq 1$$

putting in values we get:

ε_{eff} = 2.66

From this equation, we can see that the dielectric constant is small, thus overall dielectric losses in microstrip lines are smaller.

Return Loss

Return loss is related to both standing wave ratio (SWR) and reflection coefficient (Γ). Increasing return loss corresponds to lower SWR. Increasing return loss corresponds to lower SWR. Return loss is a measure of how well devices or lines are matched.

To calculate return loss, we need to find voltage reflection coefficient (Γ).

For short circuit, it is Γ = $\frac{Z_L - Z_o}{Z_L + Z_o}$

Ideally, load impedance

Z_L = 0 for short circuit.

Hence, Γ = -1

And then VSWR = $\frac{1+|\Gamma|}{1-|\Gamma|}$

VSWR = ∞

And then return loss in decibel (db) will be:

RL_{db} = -20*log10(|Γ |)

= -20*log10(|-1|) = 0db

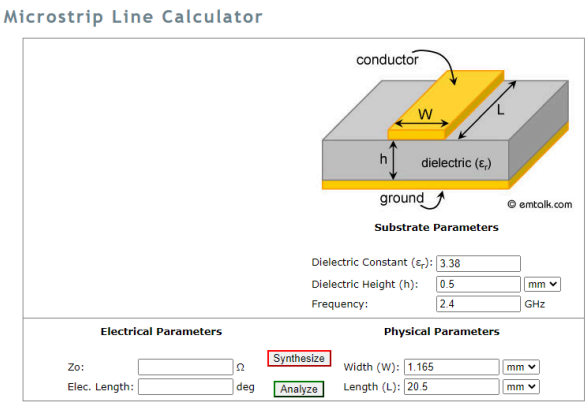
However, in practice, there will always be some amount of reflection due to imperfections in the design, fabrication of the microstrip inductor and little amount of impedance or resistance even in short circuit s line. The amount of reflection and the resulting return loss will depend on the specific design and quality of the microstrip inductor.

Verification

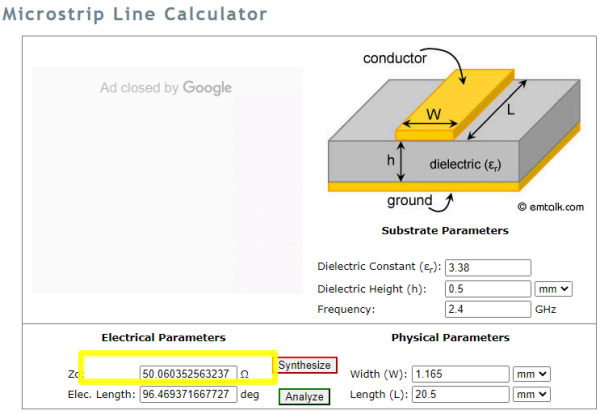
We verified our result using an online microstrip line calculator, “em:talk”.

This calculator requires an additional parameter of electrical length.

Since we do not have this parameter, we chose to input the values of length and width instead.



Entering the known parameters

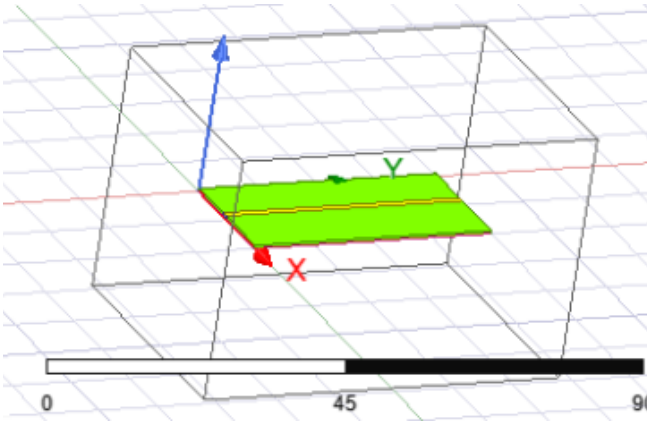


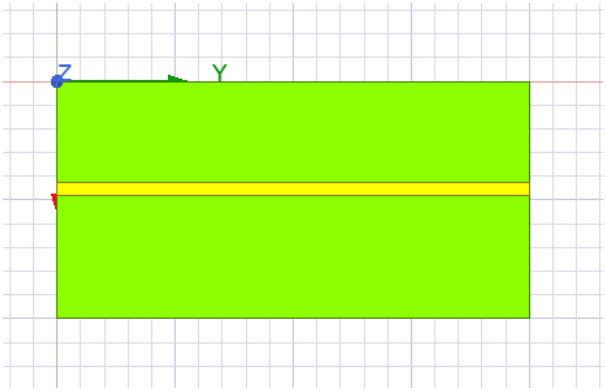
Output

Our calculated values of length and width give the impedance as 50.060 Ω which is very close to 50Ω. This validates our result.

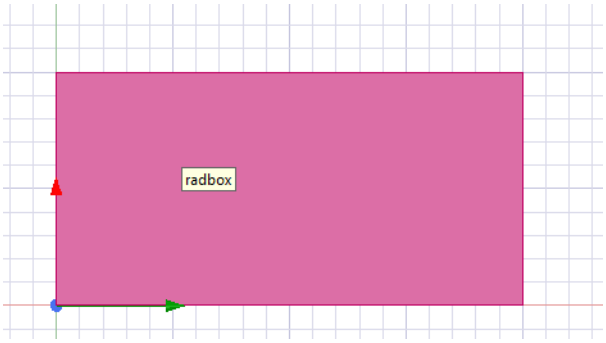
Simulation

A microstrip inductor was designed based on the data provided and calculated.



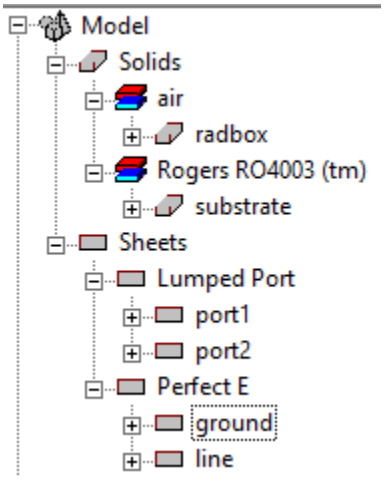


Top view

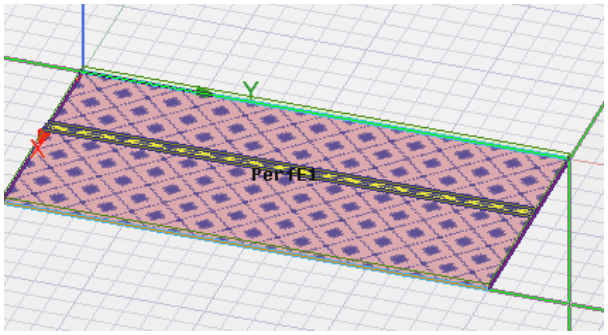


Bottom view

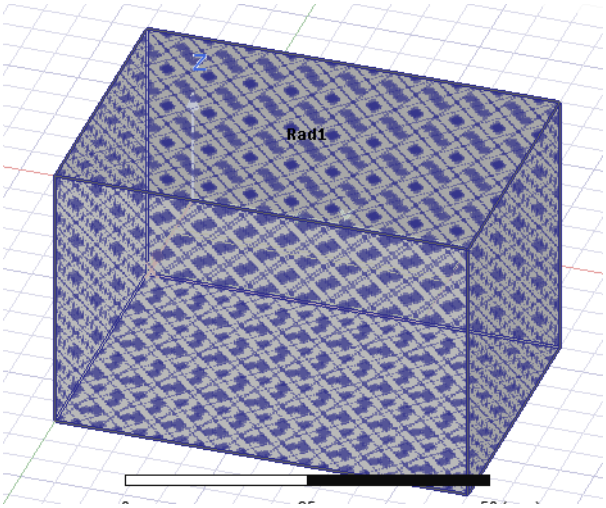
This design included:



Where ground and microstrip line were given perfect conductor(i.e: copper) boundary and radiation box was assigned radiation while port 1 and 2 were assigned excitation of a lumped port.



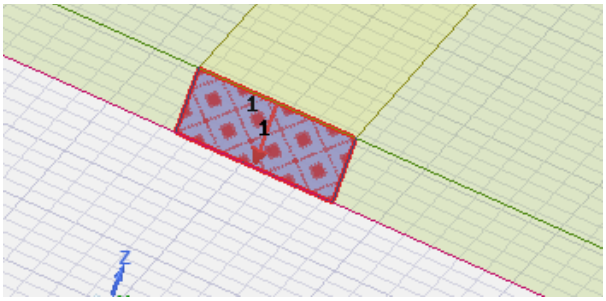
Perf E boundaries

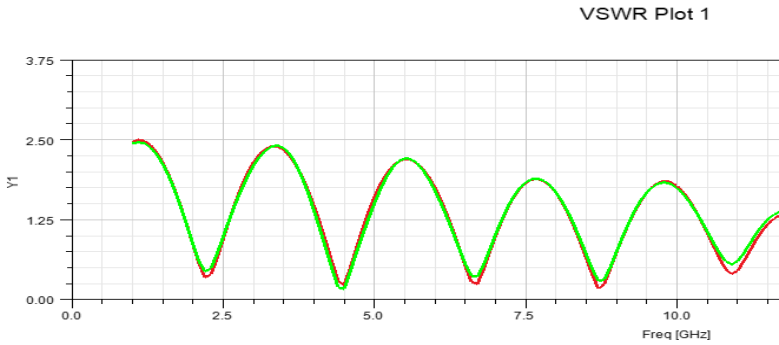


The blue line at the top indicates the return loss of the microstrip which also validates our calculation.

(Note: It is still an ideal design, hence no loss. When in reality, there is bound to be some loss due to little impedance in short circuit.)

Radiation boundary

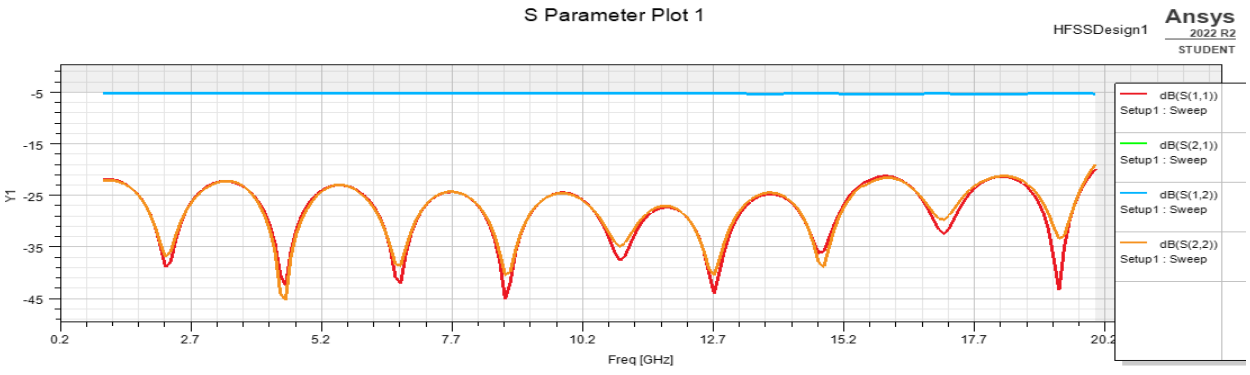




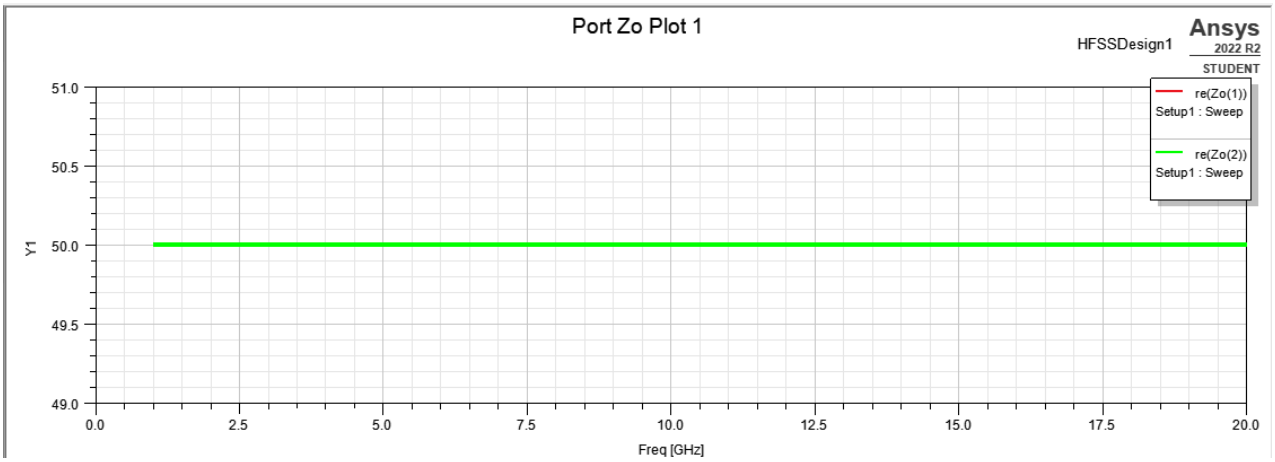
Lumped port excitation

With this design, we plotted the S parameter, impedance, and VSWR against frequency.

S-parameters describe the input-output relationship between ports (or terminals) in an electrical system.



4. Microstrip Line Calculator | em: talk (emtalk.com)



Reference

1. Fundamentals of Applied Electromagnetics by Fawwaz Tayssir **Ulaby**, Umberto **Ravaioli**.
2. How Low are Dielectric Losses in Microstrip Lines? | NWES Blog (nwengineeringllc.com)
3. 3.11: Microstrip Line - Engineering LibreTexts